IMPERIAL

High-Throughput Automated Proteomics Toward Clinical Translation

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HandyAndy

Web platform for exporting Andrew-compatible method files

→

Andrew

Liquid-handling robot executes the method automatically with high-precision pipetting

→

Prepared samples

Ready for LC-MS/MS — consistent, reproducible, and hands-free

Mass spectrometry-based proteomics sample preparation involves numerous time-consuming, labor-intensive steps that can be error prone, or suffer from high variability across batches. The Andrews+ (Waters) is a pipetting robot that mimics human-pipetting with standard lab pipettes and is controlled through a simple-to-use web-based graphical user interface.

A web-to-robot pipeline that makes liquid-handling automation practical even for small sample cohorts — reducing manual effort with improving precision.

Data Availability: All method files and descriptions are available via HandAndy:

https://harrywhitwell.shinyapps.io/HandyAndy/

Keywords: Automated sample preparation, Proteomics, HandyAndy, Andrew, LC–MS/MS

Table 1: Automated proteomics sample preparation steps used for different applications.

No.	Optimised Automated Methods	Automation Details
1	Protein Assay	Pierce Bradford Assay (CN 23200); Microplate assay
2	Protein digestion	Single-pot trypsin digestion
3	Peptide Desalting	Full automated desalting with Oasis HLB (PN 186001828BA)
4	Peptide Assay	Pierce Fluorescent Peptide Assay (23290)

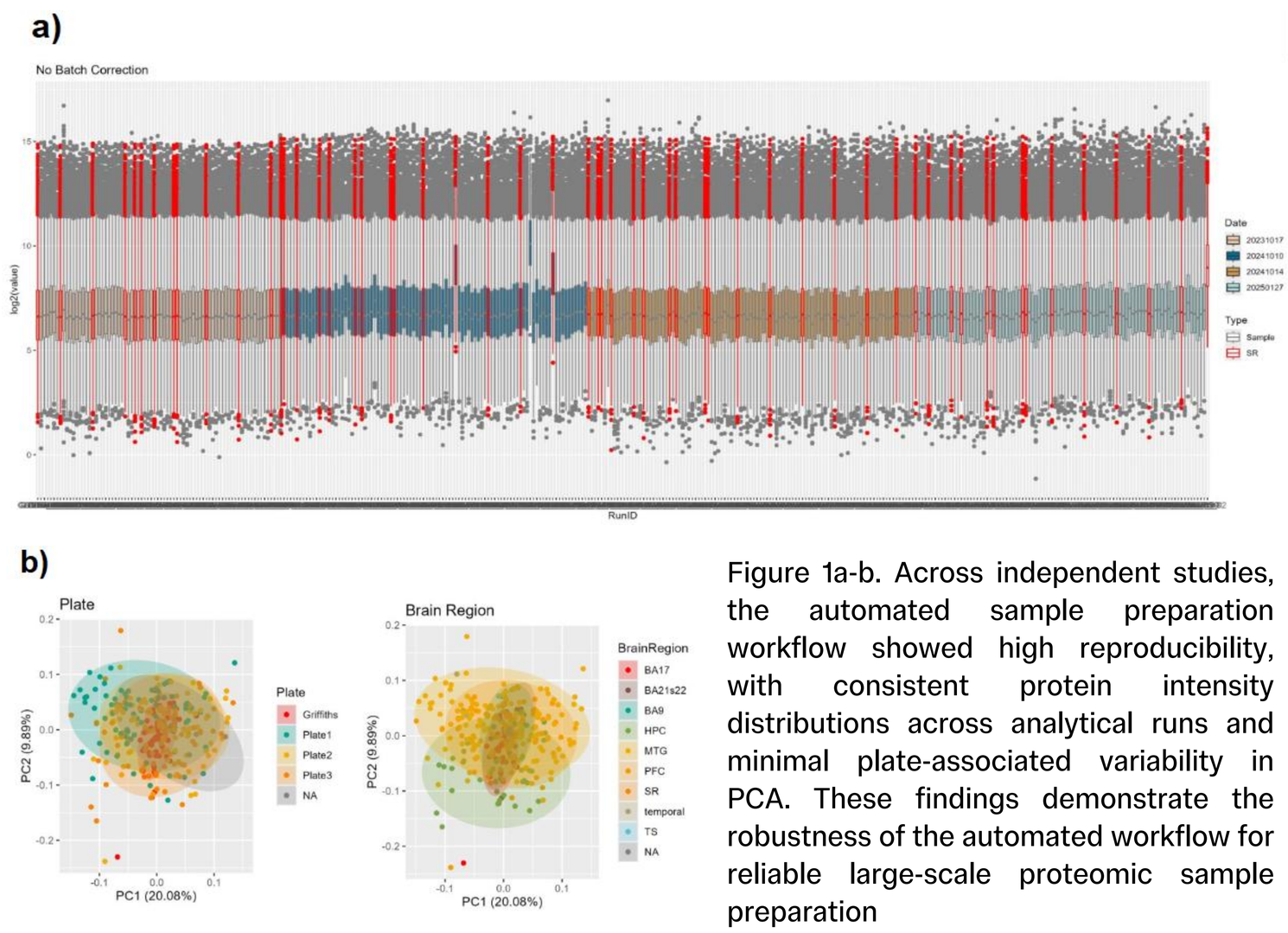
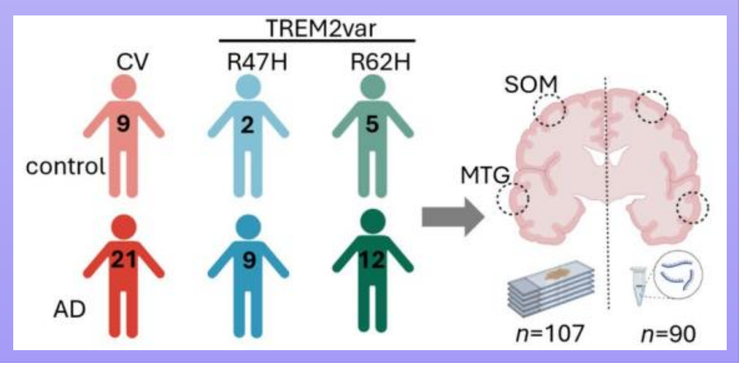


Figure 1a-b. Across independent studies, the automated sample preparation workflow showed high reproducibility, with consistent protein intensity distributions across analytical runs and minimal plate-associated variability in PCA. These findings demonstrate the robustness of the automated workflow for reliable large-scale proteomic sample preparation

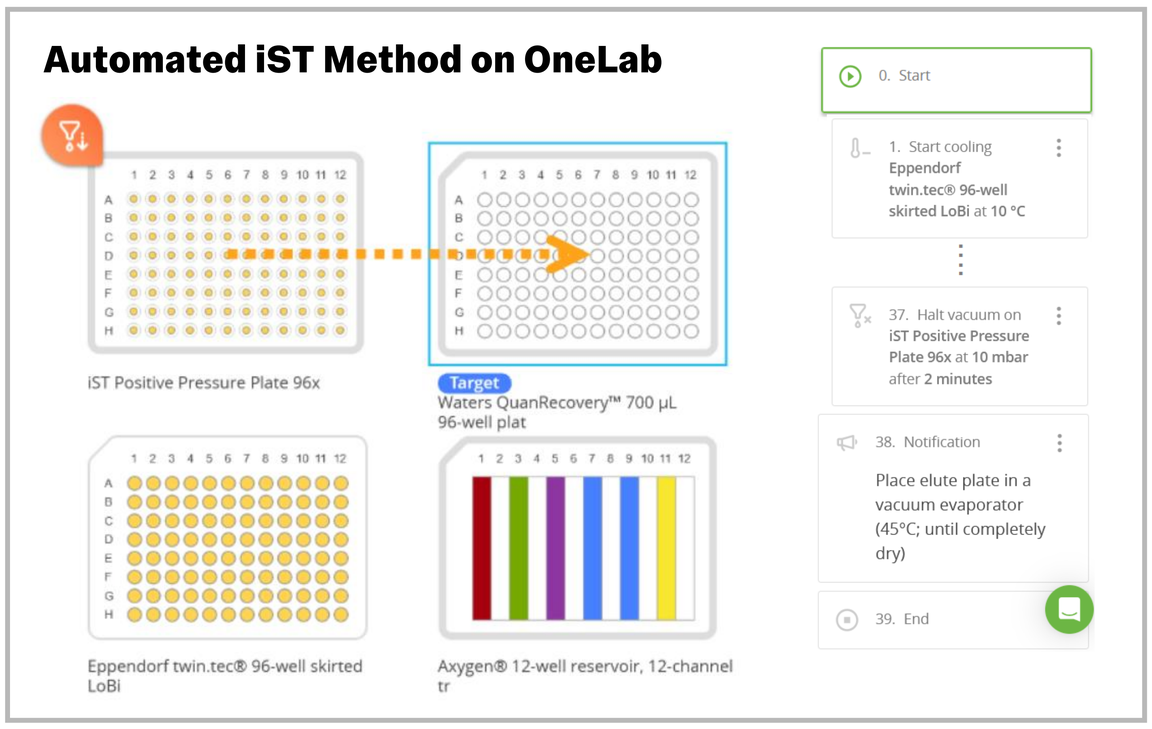
Figure 1a-b: Reproducibility of the automated sample preparation workflow across independent proteomics studies.

Application in clinical research

To ensure high-throughput reproducibility across 107 brain samples, protein digestion and desalting were automated using an Andrew+ pipetting robot (Waters) equipped with a 96-well Peltier and Vacuum+ module. This robotic workflow minimised technical variability, enabling robust large-scale Proteomic profiling to identify genotype-specific changes in Alzheimer's pathology [1].



This workflow will shortly be evaluated experimentally in our laboratory to assess its performance and further optimise the protocol under practical conditions.



PREOMICS

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Reference

1. Fancy, Nurun N., et al. "Mechanisms of increased Alzheimer's disease pathology with R47H and R62H TREM2 variants." *Acta Neuropathologica* 151.1 (2026): 67.